

Invariant Transformations and Dual-Unit Dimensional Normalization in Multi-A E-Commerce Catalogs

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Published in: Agentic Commerce Research Suite 2026

Abstract

~~Abstract: This paper introduces a novel framework for handling multi-unit dimensional data in e-commerce catalogs. It addresses the challenge of normalizing measurements across different units (e.g., grams, kilograms, ounces) and ensuring consistency across various product attributes. The proposed method leverages invariant transformations to maintain data integrity while enabling accurate comparisons and aggregations. The framework is designed to be scalable and adaptable to diverse product categories and catalog structures. Key contributions include a unified normalization process, robust handling of missing or inconsistent data, and a modular architecture for integration with existing catalog management systems. Experimental results demonstrate the effectiveness of the approach in improving data quality and enabling more precise analytics.~~